

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

An Earth observation that survives an agent handoff.

emem gives a sampled or derived observation a reusable reference. The next agent recovers its value, location, acquisition time and processing recipe, then checks the evidence.

A compact citation keeps the observation traceable across agents and time. Encode in orbit, decode in AI’s reasoning with emem.

Jaya Kumari · Avijeet Singh · Vortex AI · avijeet@vortex.ai
Agentic AI for Earth Observation · BIFOLD and ESA D-lab · Berlin · Poster Session 1 · 19 Oct 2026 · emem.dev · github.com/Vortex-AI/emem (Apache-2.0)

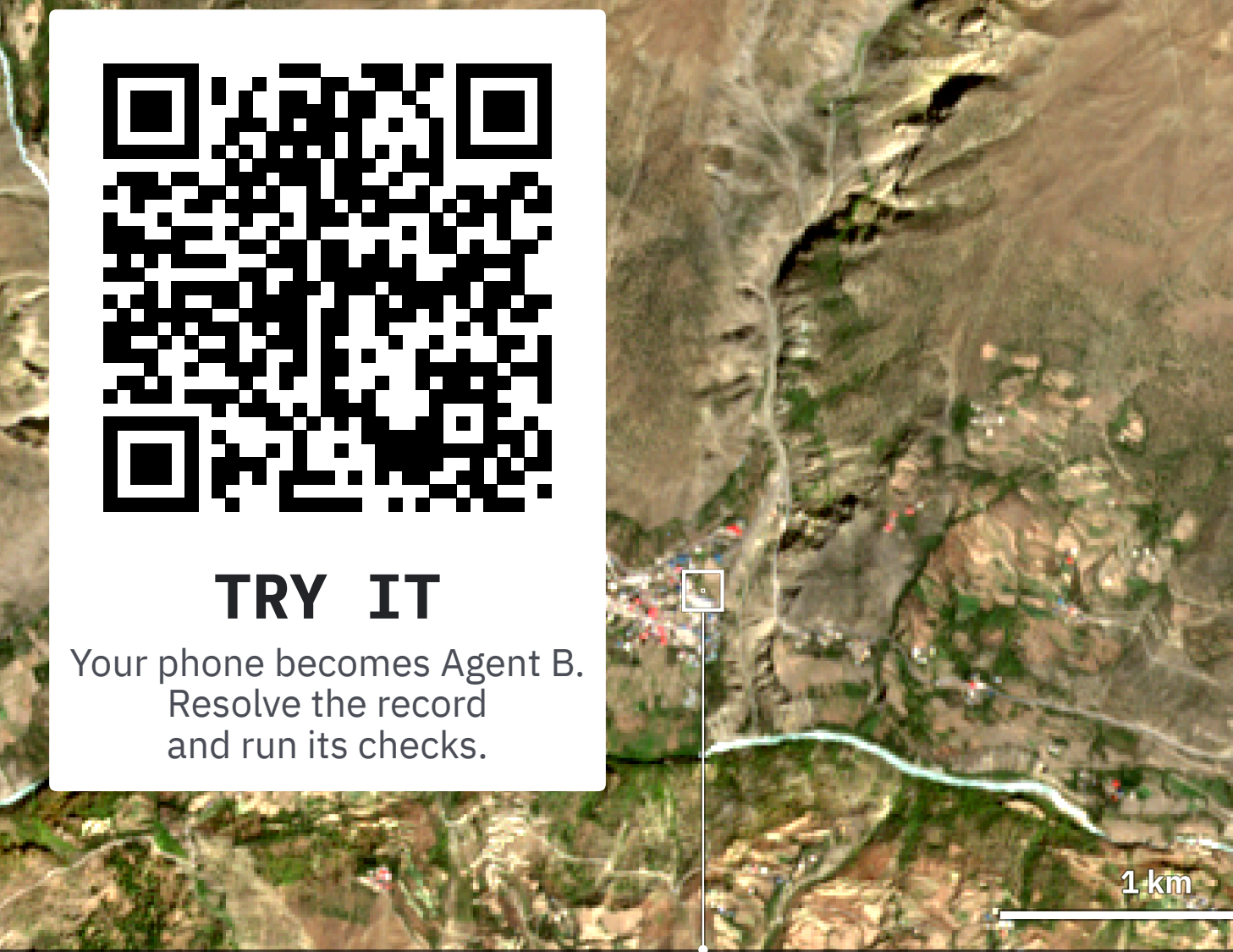
Agent A cites a satellite observation. What can agent B check without trusting A, A’s model, or us?

emem makes raw satellite observations **addressable by place, band and time**, each a signed record named by the **BLAKE3** hash of its bytes. The exact evidence A cites survives compaction, a change of model and a **handoff**; B re-hashes it, verifies the log entry and receipt offline, and traces it to the source pixel. Agents cannot write observations, and a changed, rounded or forged value no longer matches its name.

Trace: token → record bytes → cell → pixel → receipt → log → key

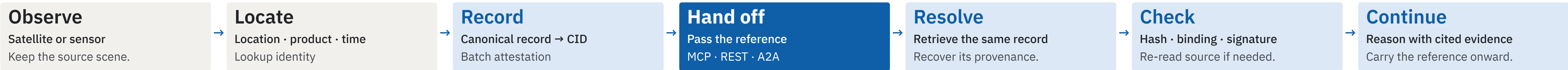


TRY IT
Your phone becomes Agent B. Resolve the record and run its checks.



2.02 GB scene → 84 characters
This pixel’s NDVI record, handed off in one text message.
84 characters, 46 tokens; one SMS holds 160.

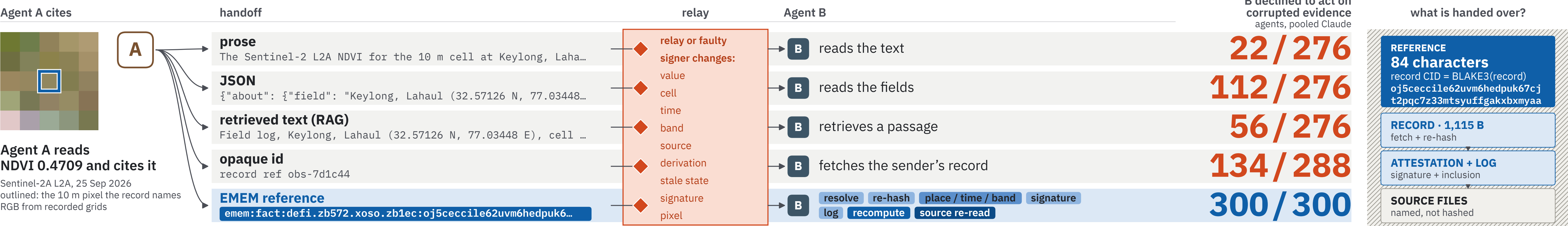
From observation to the next agent



1 What survives an agent handoff?

The same observation, carried through receiver checks.

Agent A cites a source read. A relay changes the evidence. Agent B acts on it or checks the reference.



In the controlled handoff test, agents declined 300 of 300 corruptions carried by a checked reference with explicit instructions; with prose, 22 of 276 (they acted on 254).

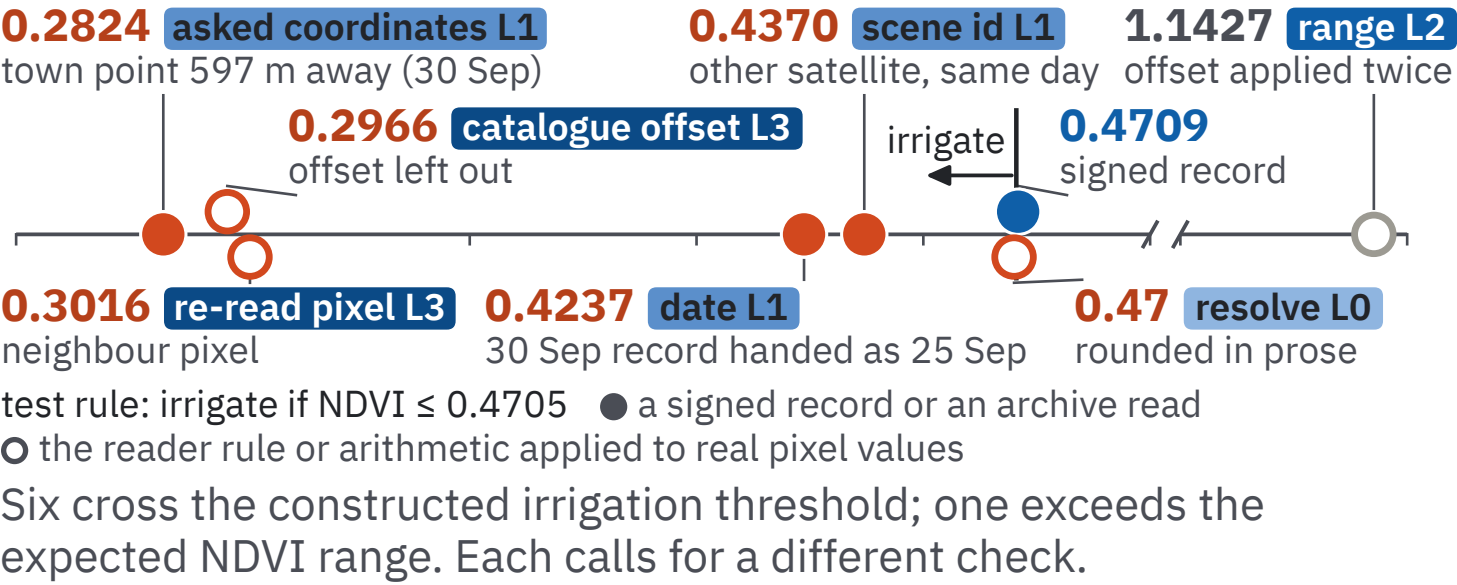
Earth data stays on the encoding device in orbit; its reference is downlinked and enters the AI’s reasoning (design; SAT-042 reference harness).

Claude Haiku 4.5, Sonnet 5.5 and Opus 5.5 as B, pooled; Qwen2.5-7B apart; 2,794 scored trials on 1 Oct 2026. Pooled results above are the Claude runs; Qwen is reported separately.
Qwen2.5-7B acted on corruptions with prose 22 of 23 times; with the checked reference, 2 of 25. One run. Simulated corruptions, constructed decision threshold; model breakdown and confidence intervals in Reproduce.

2 How does a value change between agents?

One NDVI, eight values

At Keylong on 25 Sep 2026, scene, date, pixel and arithmetic choices change the reported NDVI.



3 Where did the evidence change?

A changed value needs a cause

Between acquisitions, the surface, sensor, location or processing can change.

CONCEPTUAL CHANGE MODEL

$$\Delta Z = \Delta_{env} + \Delta_{sensor} + \Delta_{geo} + \Delta_{encoder} + \epsilon$$

Δz : change in readout. Environment · sensor · geolocation · encoder · residual. The attribution ledger links evidence to these terms.

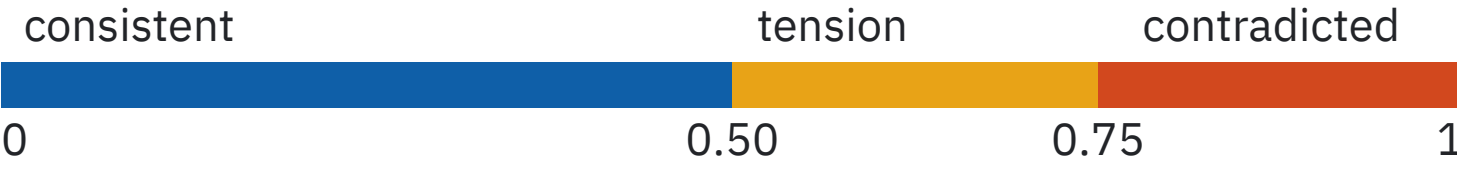
IMPLEMENTED · DRIFT-ANCHOR SCORE

$$d = |x - a| \text{ for finite } x, a \text{ and } 0 < \sigma < \infty$$

$$r = d / (3\sigma); s = r / (1 + r)$$

Invalid σ : $s = 0$ if $d = 0$; otherwise $s = 1$.

x : output · a : anchor · σ : anchor uncertainty



Tests pin $3\sigma \rightarrow 0.50$ and $9\sigma \rightarrow 0.75$.

Score boundaries: consistent < 0.50; tension < 0.75.

The implemented score measures disagreement with an anchor. SAT-042 applies it to a fixed reference in the execution harness.

7 What does a checked reference establish?

Know what was checked

CHECKABLE	NOT ESTABLISHED
Record bytes + content identity	Sensor accuracy
Declared cell, band, time	Entity identity
Attestation signature	Physical truth
Declared derivation	Decision correctness
Recorded history / as-of	Source quality: inherited

Run the corresponding checks; a content address alone is insufficient.

Record checks establish properties of the cited record. A source re-read adds evidence about the sampled pixel; sensor accuracy remains inherited.

FULL TRACE · 30 SEP 2026

Token → record → source pixel

accept if BLAKE3(b) = cid & cell(b) = cell(token) & NDVI(DN(b)) = value & COG(row, col) = DN(b) & Ed25519(o) & leaf in STH & STHgeo.qa Prefix of STH

Without emem software, a 698-line script (Python stdlib, blake3, cbor2, pynacl) checked 15 links (17 checks) in 17.9 s, all verified. Grey: what must still be trusted after each check.

- 1-3 bytes → name
- 1,115 B hash to the fact_cid; one flipped mantissa bit gives m5x2cxs...
- 4-5 cell binding hashed cell = token cell = the lat/long the derivation used · nothing
- 6-7 derivation scene id, DNs, catalogue and offset are inside the bytes; NDVI recomputed bit-for-bit · argument order read from emem’s code
- 8-9 upstream the STAC item matches the scene; the COG pixel at col 9098, row 9443 reads B08 3502, B04 1900 · STAC carries no checksum
- 10-11 server signature emem.dev’s Ed25519 receipt over the request id, time, cells and returned fact_cids · shows what emem.dev served, not that the value is right
- 12-14 public log entry 2,457,078 is in the 2,556,451-entry log, consistent with the root geo.qa co-signed · geo.qa is also run by Vortex AI
- 15 key → domain the signing key is published in did:web, JWKS and DNS TXT for emem.dev · Web PKI; DNS without DNSSEC

Links 1-5 and 12-14 fit in a 4,906-byte file that repeats 9 checks with networking disabled (unshare -rn); a 1-bit tamper, a wrong cell or a wrong key fails it.

THE CONTRIBUTION

Checkable observation handoff

An agent passes a reference to a specific physical observation; the receiver resolves it and checks the evidence.

Built with standard BLAKE3, Ed25519, CBOR and Merkle logs. STAC, openEO, PROV, C2PA, RAG and temporal storage provide complementary layers.

12 Use emem. Carry the evidence forward.

Measured separately: same record and value through 11 client paths; receipt signatures checked on 9 (30 Sep 2026).

ChatGPT
@emem plugin
Enable emem; paste a token.

Claude
Code plugin / MCP
Resolve from chat or terminal.

Visual Studio Code
MCP connector
Copilot agent mode: @mcp emem

Dify
Marketplace plugin
Referent Lock / EUDR workflows

Salesforce MuleSoft
Exchange listing
Open Vortex AI MCP Server

MORE CLIENTS
Cursor
Gemini CLI

PROTOCOLS
MCP · A2A
REST / OpenAPI

DEVELOPER
Python · TypeScript
Docker · LangGraph memory
GitHub source

FRAMEWORK EXAMPLES
LangChain · LlamaIndex
AutoGen · CrewAI · Mastra
Agno (see setup notes)

DISCOVERY / MIRRORS
GitHub MCP · MCP Registry
Glama · Smithery · ClawHub
Context7 · Hugging Face (older mirror)



Setup · EO tasks
Reproduce